

Problem Statement No 20 - AI Agent for Electricity Consumption Awareness

The Challenge

Many consumers lack awareness about how daily activities and appliance usage impact their electricity consumption. This leads to inefficient usage and unnecessary energy wastage. The challenge is to build an AI Agent that educates users about electricity consumption patterns and promotes energy-efficient behavior.

The Objective

1. Energy Consumption Analysis :-The AI agent should analyze household electricity usage patterns and identify areas where energy consumption is unusually high.
2. User Awareness and Education :- The system should provide simple explanations that help users understand how their daily activities affect electricity consumption.
3. Energy Saving Suggestions :- The AI agent should provide practical tips and recommendations that help users reduce electricity usage and adopt energy-efficient practices.
4. Conversational Interaction :- Users should be able to ask questions related to electricity usage and receive helpful insights and suggestions from the system.

Tech Stack (Mandatory)

LangFlow + IBM watsonx.ai + Granite Models

IBM Cloud Agentic Lab with IBM Granite Models

Replicate with LangChain + RAG + IBM Granite Models